(https://databricks.com)

Lab Solution: Querying Files

Run the following cell to setup the lab environment

```
4
%run ../Includes/Setup-Lab
```

Q1- Extracting data directly from Parquet files

Use a SELECT statement to directly query the content of the Parquet files in the directory **${dataset.school}/enrollments**

```
6
-- ANSWER
SELECT * FROM parquet.`${dataset.school}/enrollments`
```

Use the above SELECT query in a CTAS statement to create the table **enrollments**

```
8
-- ANSWER
CREATE TABLE enrollments AS
SELECT * FROM parquet.`${dataset.school}/enrollments`
```

Run the below cell to ensure data was written as expected in the **enrollments** table

```
10
SELECT * FROM enrollments
```

Q2- Registering Tables from JSON Files

Use CTAS statement to create the table **students** from the json files in the directory: **${dataset.school}/students-json**

```
12
-- ANSWER
CREATE TABLE students AS
SELECT * FROM json.`${dataset.school}/students-json`;
```

Run the below cell to ensure data was written as expected in the **students** table

```
14
SELECT * FROM students
```

Q3- Registering Tables from CSV Files

Create the temporary view **courses_tmp_vw** from the csv files in the directory: **${dataset.school}/courses-csv**

Knowing that:
- The delimiter is semicolon (**;**)
- There is a header of column names in each file

The schema for the view:

Column Name	Column Type
course_id	STRING
title	STRING
instructor	STRING
category	STRING
price	DOUBLE

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```
-- ANSWER
CREATE TEMP VIEW courses_tmp_vw
   (course_id STRING, title STRING, instructor STRING, category STRING, price DOUBLE)
USING CSV
OPTIONS (
  path = "${dataset.school}/courses-csv",
  header = "true",
  delimiter = ";"
);
```

Create the manged table **courses** from the temporary view **courses_tmp_vw**

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```
-- ANSWER
CREATE TABLE courses AS
  SELECT * FROM courses_tmp_vw;
```

Query the data in the **courses** table to ensure data was written as expected.

20

```
-- ANSWER
SELECT * FROM courses
```

Finally, review the metadata information of the table **courses**, and verify that the table type is Managed

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```
DESCRIBE EXTENDED courses
```

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